Windows Security Log Analysis

Lab 1 Evidence and Investigation Report

SOC Analyst Practice Hub

Offline analysis of the repository-provided synthetic Windows Security event dataset, with supporting PowerShell and Event Viewer evidence.

Report scope	Value
Dataset	windows-security-events.csv
Environment	Authorized Windows lab workflow and offline analyzer
Time range (UTC)	2026-11-03T05:30:00Z to 2026-11-03T05:47:00Z
Records analyzed	12
Evidence images	25 cataloged screenshots

Important: The first 23 screenshots document the original exercise sequence and contain redacted copies of source material. The final two are explicitly synthetic demonstrations. This report does not treat any screenshot as proof of compromise.

1. Executive summary

The offline dataset contains 12 synthetic Windows Security-style events from 2026-11-03T05:30:00Z through 2026-11-03T05:47:00Z. The main investigative lead is a burst of 6 failed logons for **alex.lee** from **203.0.113.42**, followed by a successful logon for the same account and source. This pattern warrants correlation, but it does not by itself establish brute force, unauthorized access, or compromise.

The dataset also includes explicit-credential use, network-share access, account creation, local-group membership change, and special-privilege assignment. These may be routine administration in an authorized lab. Validate them against change records, endpoint telemetry, authentication context, and owner approval before escalating.

Assessment: investigative lead only. **Confidence:** high for the observations, low for incident attribution because the dataset is synthetic and intentionally limited.

2. Scope, sources, and limitations

Item	Details
Scope	Offline review of the bundled CSV plus the documented PowerShell/Event Viewer workflow.
Data source	data/windows-security-events.csv; all rows are labeled synthetic training events.
Analyzer	tools/Analyze-SecurityEvents.ps1; reference window is relative to the newest dataset event.
Missing context	Full Windows XML fields, process IDs, status/substatus, endpoint/network telemetry, authorization records, and change tickets.
Safety boundary	No live Security log query or event generation is required for the offline exercise.

3. Key findings

Finding	Observed result	Interpretation
Failed logons (4625)	6	Short burst against one account; lead for correlation, not proof of brute force.
Successful logons (4624)	1	A success follows the failures; validate account ownership and session context.
Top failed-logon source	203.0.113.42 (6)	Documentation-only address in synthetic data.
Targeted account	alex.lee (6)	Confirm whether the account and activity were expected.
Other review events	4648, 4672, 4720, 4732, 5140	Review credentials, privileges, account/group changes, and share access.

4. UTC timeline

Timestamp	Event	Account / source	Action	Interpretation
2026-11-03T05:30:00Z	4625	alex.lee / 203.0.113.42	failed_logon	Failed-logon burst
2026-11-03T05:31:00Z	4625	alex.lee / 203.0.113.42	failed_logon	Failed-logon burst
2026-11-03T05:32:00Z	4625	alex.lee / 203.0.113.42	failed_logon	Failed-logon burst
2026-11-03T05:33:00Z	4625	alex.lee / 203.0.113.42	failed_logon	Failed-logon burst
2026-11-03T05:34:00Z	4625	alex.lee / 203.0.113.42	failed_logon	Failed-logon burst

Timestamp	Event	Account / source	Action	Interpretation
2026-11-03T05:35:00Z	4625	alex.lee / 203.0.113.42	failed_logon	Failed-logon burst
2026-11-03T05:37:00Z	4624	alex.lee / 203.0.113.42	successful_logon	Success after failures
2026-11-03T05:40:00Z	4648	svc.backup / 192.0.2.15	explicit_credentials	Review in administrative context
2026-11-03T05:42:00Z	5140	svc.backup / 192.0.2.15	network_share_access	Review in administrative context
2026-11-03T05:45:00Z	4720	helpdesk.lab / 192.0.2.20	user_created	Review in administrative context
2026-11-03T05:46:00Z	4732	helpdesk.lab / 192.0.2.20	local_group_change	Review in administrative context
2026-11-03T05:47:00Z	4672	admin.lab / 192.0.2.20	special_privileges	Review in administrative context

5. Method and reproducibility

The lab teaches read-only PowerShell queries for recent Security events, Event ID filtering, UTC/time-window review, named XML field extraction, grouping by source/account/logon type, and evidence export. The offline analyzer validates the CSV schema, parses ISO 8601 timestamps, derives a window from the newest event, counts 4625 and 4624 events, and groups review-worthy event IDs.

Reproduction command: .\tools\Analyze-SecurityEvents.ps1 -InputPath .\data\windows-security-events.csv -OutputPath .\output\windows-security-analysis.md

Use named event fields rather than fixed Properties[] indexes because event schemas can vary by event ID and Windows version.

6. Alternative explanations and limitations

Possible benign explanations include authorized RDP testing, a mistyped password, scheduled service activity, helpdesk account administration, backup operations, and approved group membership changes. Source addresses and usernames in this synthetic dataset do not establish attribution. The CSV omits the richer XML and process/network context available in a full Windows event record.

Do not label the activity as an incident solely from repeated failures or a success after failures. Preserve the relevant event context and obtain owner confirmation before blocking, disabling accounts, or changing system state.

7. Recommendations and disposition

Priority	Action
1	Preserve related Windows event records and exact UTC timestamps; retain account, host, source, logon type, status/substatus, process, and session fields where available.
2	Correlate failed and successful logons with endpoint, RDP/VPN, firewall, application, and authentication telemetry.
3	Confirm whether account creation, local-group membership, explicit credentials, share access, and special privileges were approved administrative actions.
4	If activity is unauthorized after correlation, follow the approved containment and incident-response process.
Disposition	Monitor / validate context; do not escalate on the synthetic dataset alone.

8. Conclusion

This lab demonstrates a disciplined Windows Security log review using read-only queries, named XML fields, an offline analyzer, and documented evidence. The synthetic dataset shows six failed logons against alex.lee from 203.0.113.42, followed two minutes later by a successful logon for the same account and source. That sequence is important to investigate, but it is not sufficient to conclude brute force or compromise.

The remaining events demonstrate why analysts must correlate authentication activity with account creation, group membership, special privileges, explicit credentials, and network-share access. The correct disposition for this training dataset is **monitor and validate context**, not automatic containment.

Before escalating a real-world case, preserve the original event records, confirm the UTC timeline, identify the responsible account and host, correlate endpoint and network telemetry, and obtain owner approval. All conclusions in this report are limited by the synthetic dataset and its missing Windows event context.

9. Evidence catalog

Each evidence page below contains one screenshot and its documented purpose. The first 23 images are redacted copies of source training material; the final two are synthetic demonstrations. Review the screenshot guide before publishing or sharing evidence from any real environment.

Evidence 01: 01-powershell-version.png

```
Windows PowerShell 5.1
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Users[redacted]  $PSVersionTable.PSVersion

Major  Minor  Build  Revision
-----  -----  -----  --------
5      1      26100  9818
```

What it shows: PowerShell version check for the authorized Windows lab workflow.

Explanation: This confirms the PowerShell version and command-line environment used for the read-only Security log exercises. Recording the tool version improves reproducibility and helps explain differences in command behavior between systems.

Evidence 02: 02-recent-security-events.png

```
[redacted]et-ExecutionPolicy -ExecutionPolicy RemoteSigned -Scope CurrentUser
[redacted]et-WinEvent -LogName Security -MaxEvents 10

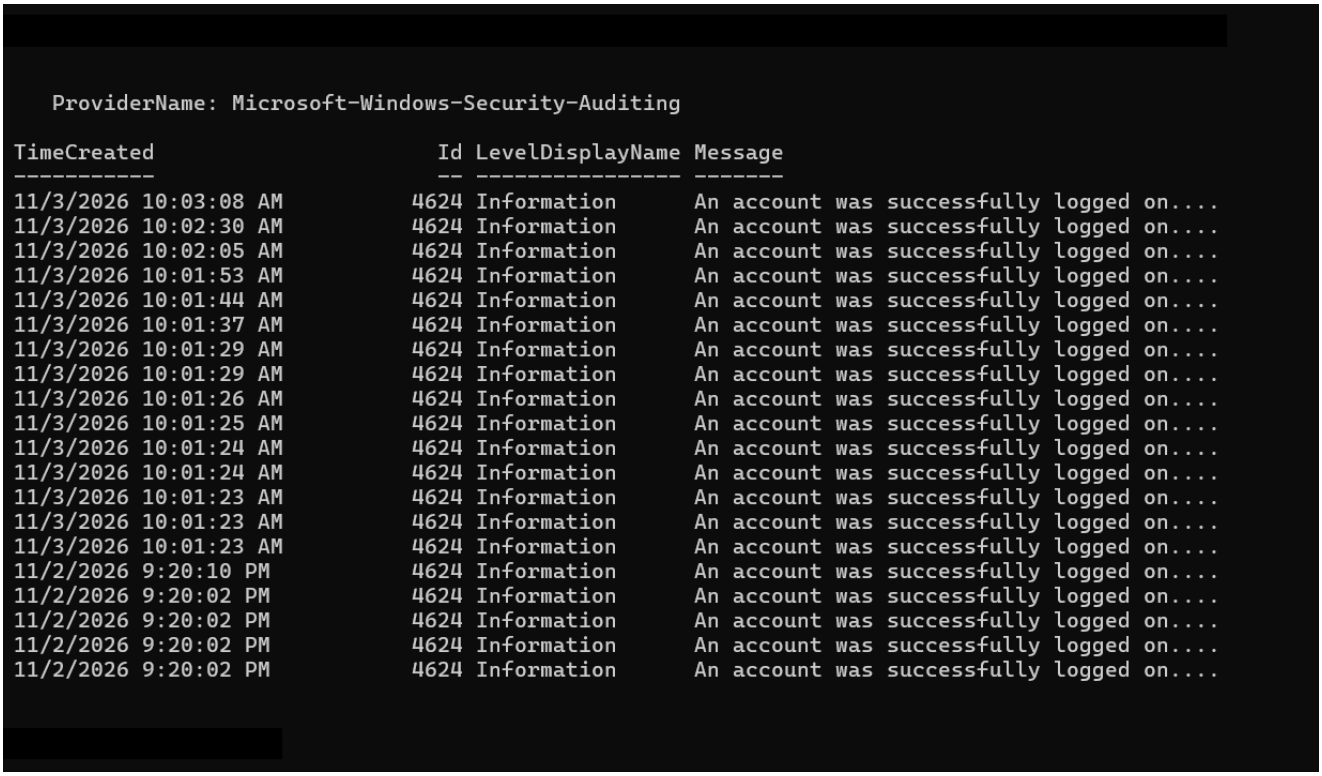

   ProviderName: Microsoft-Windows-Security-Auditing

TimeCreated                     Id LevelDisplayName Message
-----------                     -- ---------------- -------
11/3/2026 10:04:43 AM         5061 Information      Cryptographic operation....
11/3/2026 10:04:43 AM         5058 Information      Key file operation....
11/3/2026 10:04:39 AM         5061 Information      Cryptographic operation....
11/3/2026 10:04:39 AM         5058 Information      Key file operation....
11/3/2026 10:04:38 AM         5061 Information      Cryptographic operation....
11/3/2026 10:04:38 AM         5058 Information      Key file operation....
11/3/2026 10:04:33 AM         5379 Information      Credential Manager credentials were read....
11/3/2026 10:04:33 AM         5379 Information      Credential Manager credentials were read....
11/3/2026 10:04:33 AM         5379 Information      Credential Manager credentials were read....
11/3/2026 10:04:33 AM         5379 Information      Credential Manager credentials were read....
```

What it shows: Query for the ten most recent Windows Security events.

Explanation: This query provides an initial view of the newest Security events before narrower filters are applied. The analyst can confirm that the log is readable, identify the available event types, and establish a starting point for the investigation.

Evidence 03: 03-filter-successful-logons.png



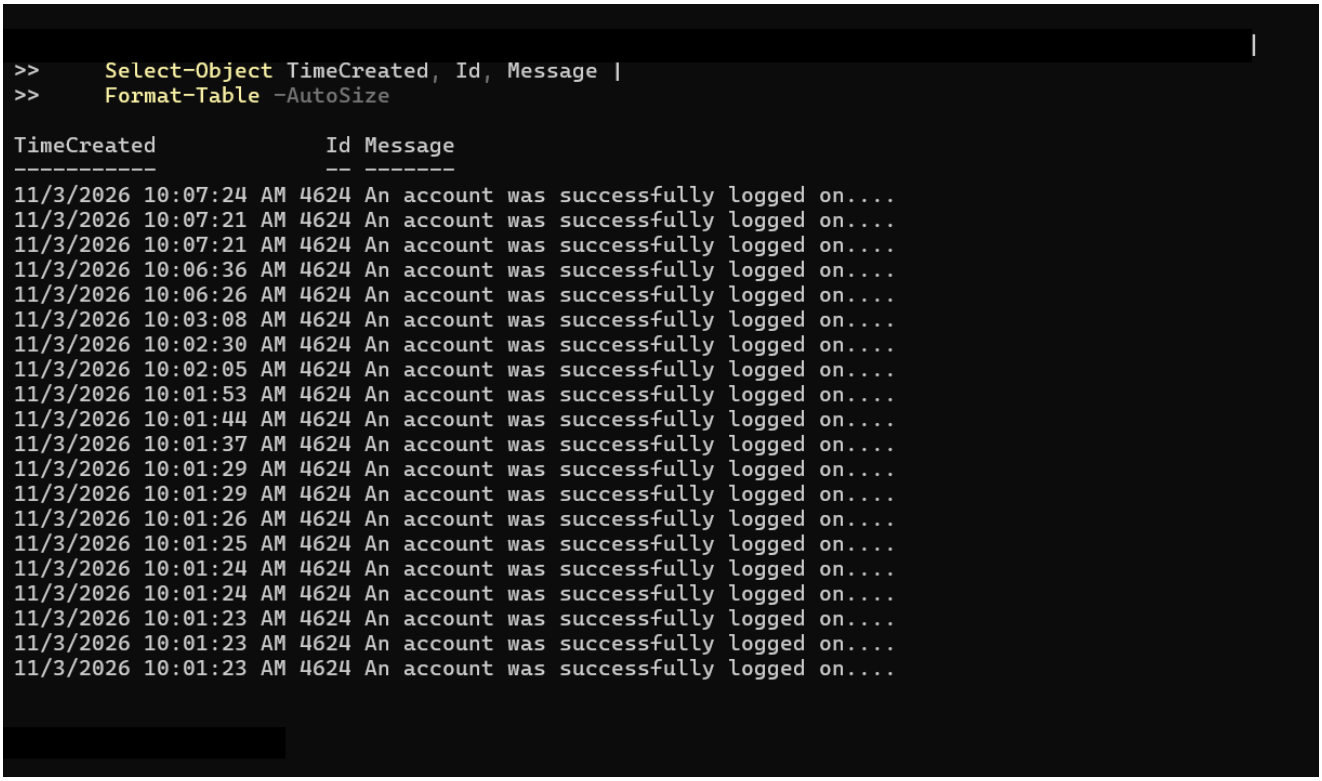
```
   ProviderName: Microsoft-Windows-Security-Auditing

TimeCreated                     Id LevelDisplayName Message
-----------                     -- ---------------- -------
11/3/2026 10:03:08 AM         4624 Information      An account was successfully logged on....
11/3/2026 10:02:30 AM         4624 Information      An account was successfully logged on....
11/3/2026 10:02:05 AM         4624 Information      An account was successfully logged on....
11/3/2026 10:01:53 AM         4624 Information      An account was successfully logged on....
11/3/2026 10:01:44 AM         4624 Information      An account was successfully logged on....
11/3/2026 10:01:37 AM         4624 Information      An account was successfully logged on....
11/3/2026 10:01:29 AM         4624 Information      An account was successfully logged on....
11/3/2026 10:01:29 AM         4624 Information      An account was successfully logged on....
11/3/2026 10:01:26 AM         4624 Information      An account was successfully logged on....
11/3/2026 10:01:25 AM         4624 Information      An account was successfully logged on....
11/3/2026 10:01:24 AM         4624 Information      An account was successfully logged on....
11/3/2026 10:01:24 AM         4624 Information      An account was successfully logged on....
11/3/2026 10:01:23 AM         4624 Information      An account was successfully logged on....
11/3/2026 10:01:23 AM         4624 Information      An account was successfully logged on....
11/3/2026 10:01:23 AM         4624 Information      An account was successfully logged on....
11/2/2026 9:20:10 PM          4624 Information      An account was successfully logged on....
11/2/2026 9:20:02 PM          4624 Information      An account was successfully logged on....
11/2/2026 9:20:02 PM          4624 Information      An account was successfully logged on....
11/2/2026 9:20:02 PM          4624 Information      An account was successfully logged on....
11/2/2026 9:20:02 PM          4624 Information      An account was successfully logged on....
```

What it shows: Filter for successful logons using Event ID 4624.

Explanation: Event ID 4624 identifies successful logons and is useful for checking whether access occurred after failed attempts. Account, source, host, logon type, timestamp, and session details should be retained for correlation; a successful logon is not automatically malicious.

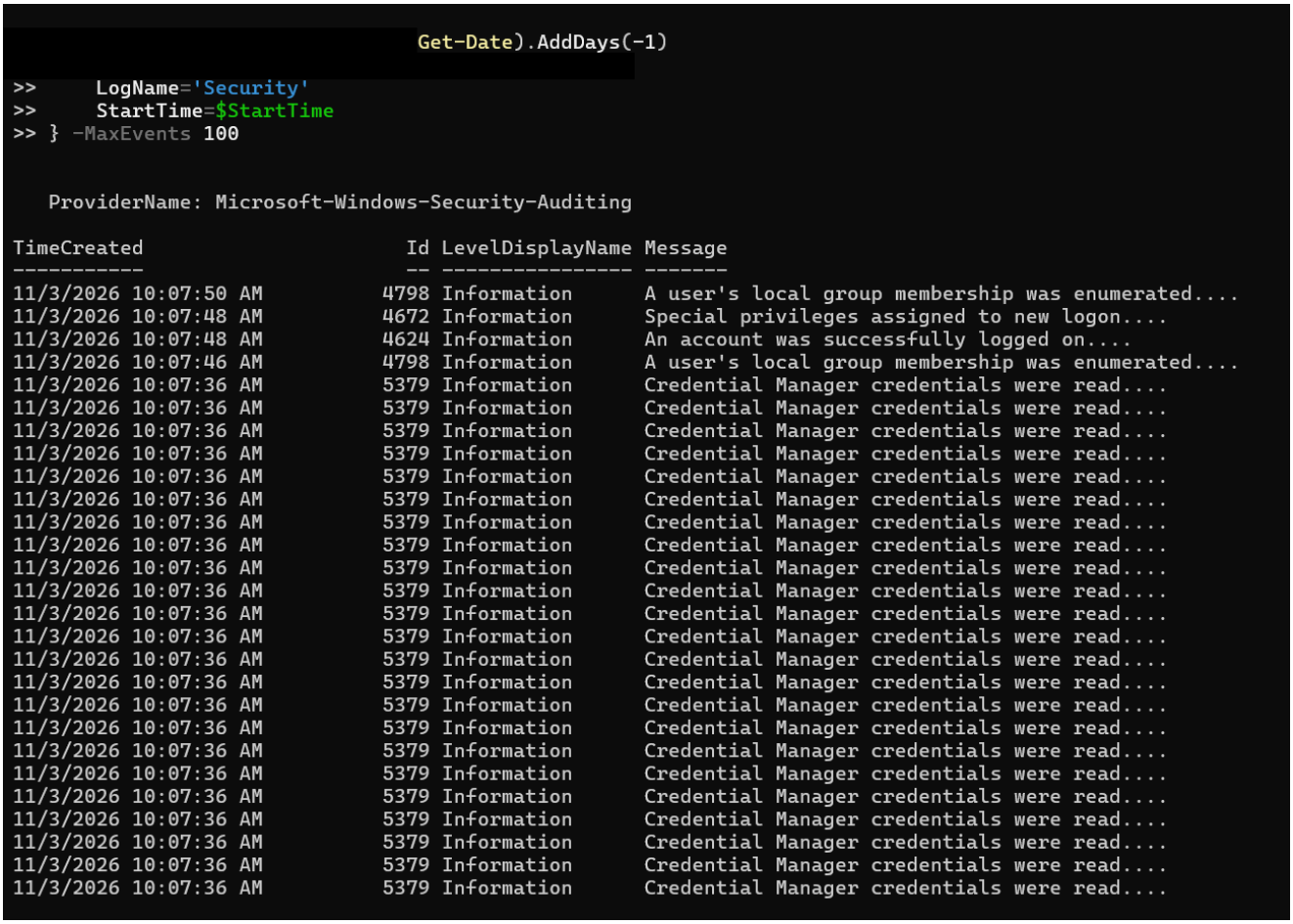
Evidence 04: 04-format-successful-logons.png



What it shows: Display selected successful-logon fields in a table.

Explanation: Formatting selected named fields makes successful-logon records easier to compare across time, account, and source. Using field names is safer than relying on fixed Properties[] positions because event schemas can vary by Windows version and event revision.

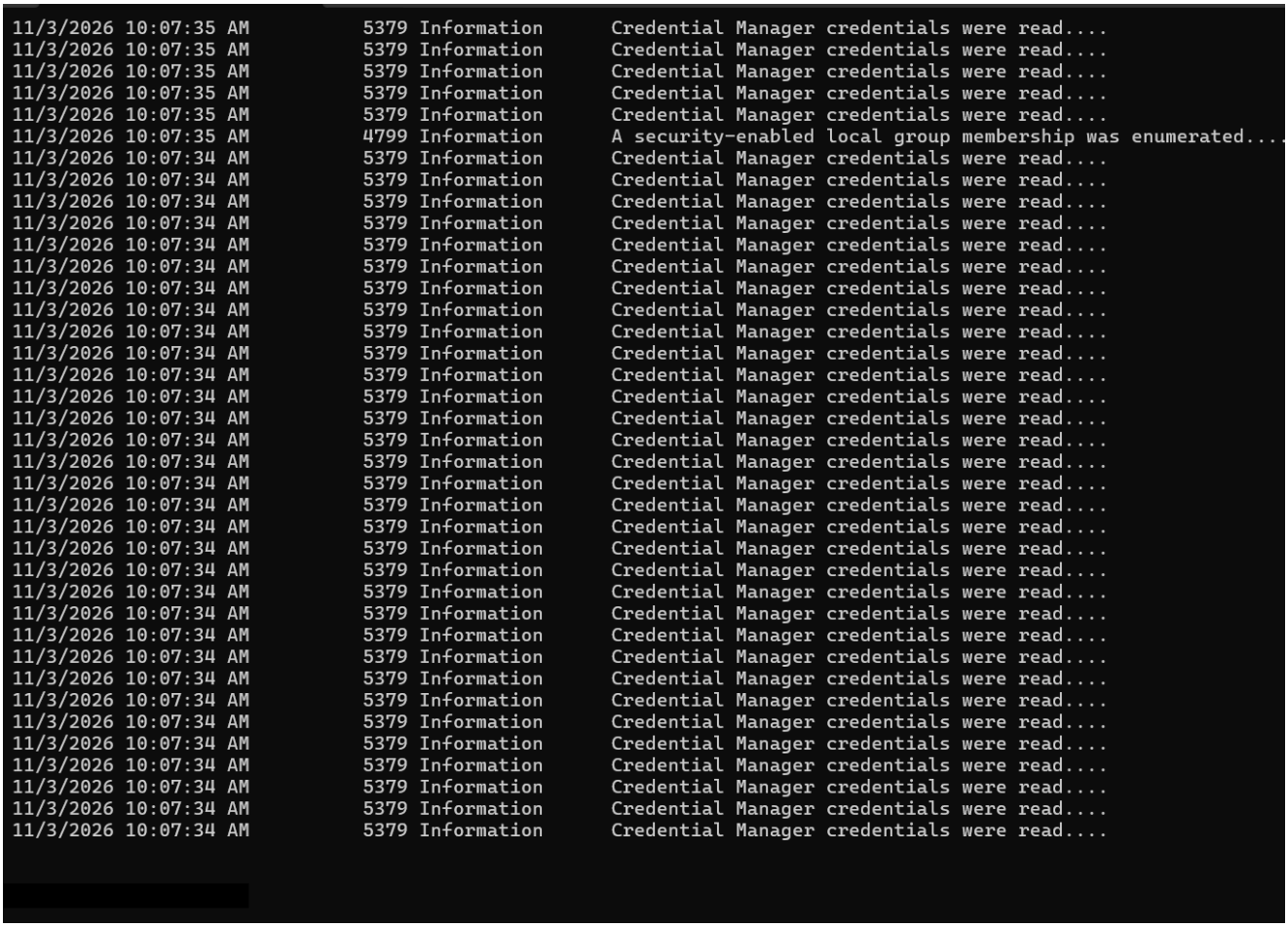
Evidence 05: 05-filter-security-events-time-range.png



What it shows: Query Security events from the previous 24 hours.

Explanation: A bounded time window limits unrelated records and makes the query easier to reproduce. Analysts should confirm the timezone and exact start time before comparing results with endpoint, VPN, RDP, or application telemetry.

Evidence 06: 06-time-range-event-results.png

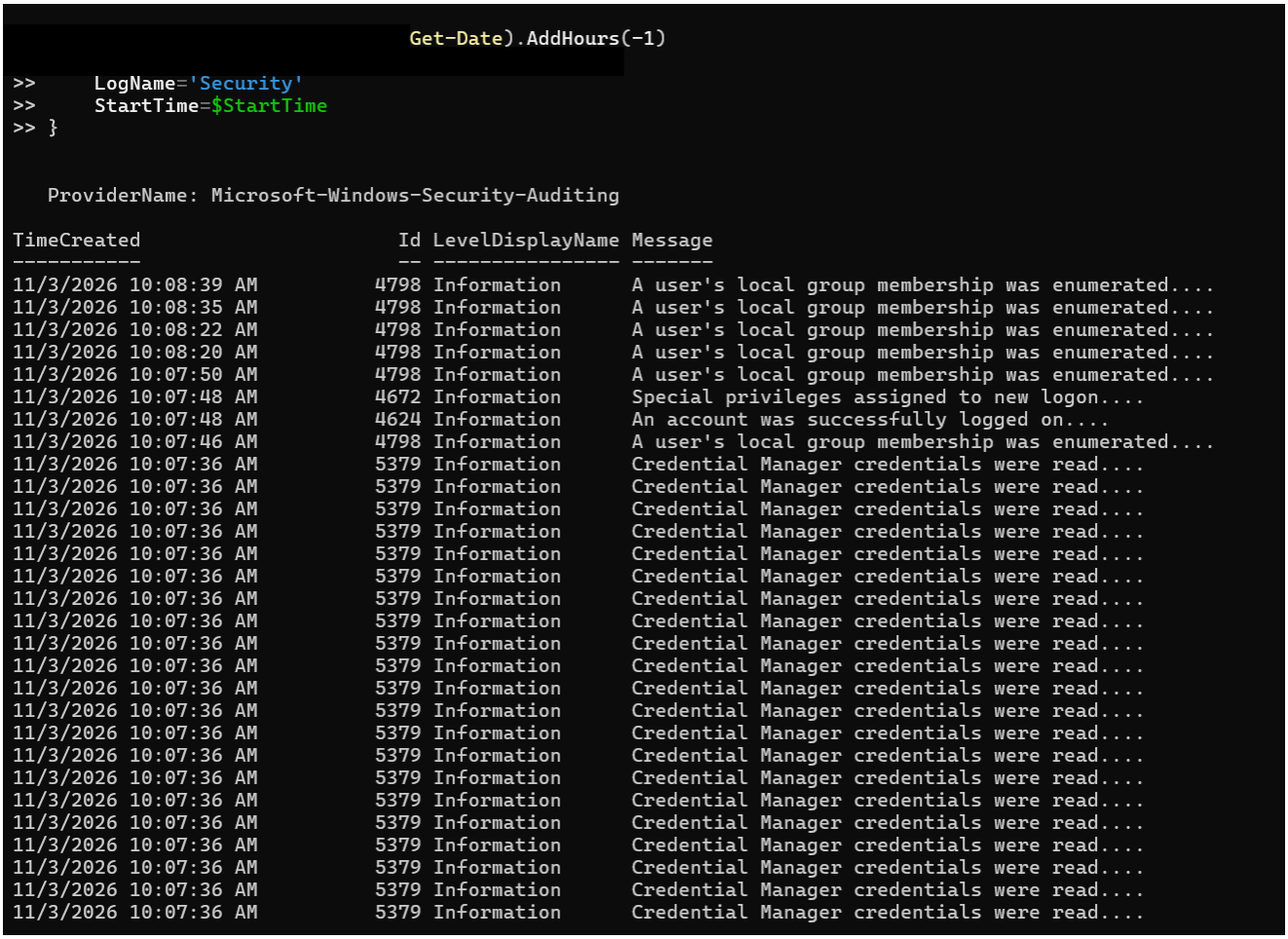


```
11/3/2026 10:07:35 AM          5379 Information      Credential Manager credentials were read....
11/3/2026 10:07:35 AM          5379 Information      Credential Manager credentials were read....
11/3/2026 10:07:35 AM          5379 Information      Credential Manager credentials were read....
11/3/2026 10:07:35 AM          5379 Information      Credential Manager credentials were read....
11/3/2026 10:07:35 AM          5379 Information      Credential Manager credentials were read....
11/3/2026 10:07:35 AM          4799 Information      A security-enabled local group membership was enumerated....
11/3/2026 10:07:34 AM          5379 Information      Credential Manager credentials were read....
11/3/2026 10:07:34 AM          5379 Information      Credential Manager credentials were read....
11/3/2026 10:07:34 AM          5379 Information      Credential Manager credentials were read....
11/3/2026 10:07:34 AM          5379 Information      Credential Manager credentials were read....
11/3/2026 10:07:34 AM          5379 Information      Credential Manager credentials were read....
11/3/2026 10:07:34 AM          5379 Information      Credential Manager credentials were read....
11/3/2026 10:07:34 AM          5379 Information      Credential Manager credentials were read....
11/3/2026 10:07:34 AM          5379 Information      Credential Manager credentials were read....
11/3/2026 10:07:34 AM          5379 Information      Credential Manager credentials were read....
11/3/2026 10:07:34 AM          5379 Information      Credential Manager credentials were read....
11/3/2026 10:07:34 AM          5379 Information      Credential Manager credentials were read....
11/3/2026 10:07:34 AM          5379 Information      Credential Manager credentials were read....
11/3/2026 10:07:34 AM          5379 Information      Credential Manager credentials were read....
11/3/2026 10:07:34 AM          5379 Information      Credential Manager credentials were read....
11/3/2026 10:07:34 AM          5379 Information      Credential Manager credentials were read....
11/3/2026 10:07:34 AM          5379 Information      Credential Manager credentials were read....
11/3/2026 10:07:34 AM          5379 Information      Credential Manager credentials were read....
11/3/2026 10:07:34 AM          5379 Information      Credential Manager credentials were read....
11/3/2026 10:07:34 AM          5379 Information      Credential Manager credentials were read....
11/3/2026 10:07:34 AM          5379 Information      Credential Manager credentials were read....
11/3/2026 10:07:34 AM          5379 Information      Credential Manager credentials were read....
11/3/2026 10:07:34 AM          5379 Information      Credential Manager credentials were read....
11/3/2026 10:07:34 AM          5379 Information      Credential Manager credentials were read....
11/3/2026 10:07:34 AM          5379 Information      Credential Manager credentials were read....
11/3/2026 10:07:34 AM          5379 Information      Credential Manager credentials were read....
11/3/2026 10:07:34 AM          5379 Information      Credential Manager credentials were read....
11/3/2026 10:07:34 AM          5379 Information      Credential Manager credentials were read....
11/3/2026 10:07:34 AM          5379 Information      Credential Manager credentials were read....
11/3/2026 10:07:34 AM          5379 Information      Credential Manager credentials were read....
11/3/2026 10:07:34 AM          5379 Information      Credential Manager credentials were read....
11/3/2026 10:07:34 AM          5379 Information      Credential Manager credentials were read....
11/3/2026 10:07:34 AM          5379 Information      Credential Manager credentials were read....
11/3/2026 10:07:34 AM          5379 Information      Credential Manager credentials were read....
11/3/2026 10:07:34 AM          5379 Information      Credential Manager credentials were read....
```

What it shows: Sample results returned by the time-range query.

Explanation: These returned records demonstrate what the selected time range actually contains and provide context for later event filtering. The result should be treated as an evidence sample, not as a complete incident conclusion without additional sources and surrounding events.

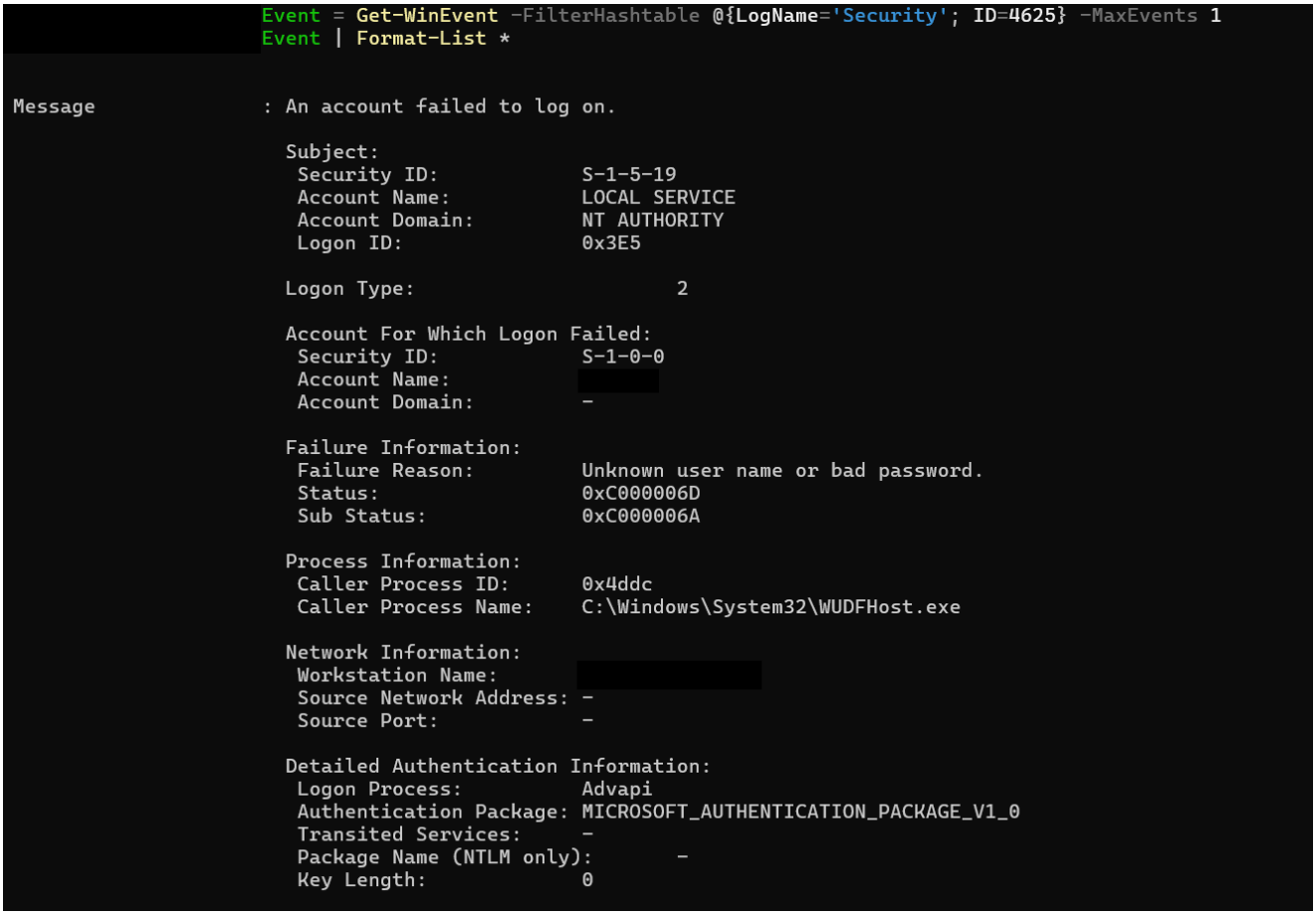
Evidence 07: 07-last-hour-event-results.png



What it shows: Security events returned by the one-hour query.

Explanation: The shorter window focuses review on the most recent activity and can reveal event order around a suspected sequence. Event age alone does not establish priority or malicious intent, so the analyst should preserve the full relevant timeline.

Evidence 08: 08-failed-logon-event-details.png



```
Event = Get-WinEvent -FilterHashtable @{LogName='Security'; ID=4625} -MaxEvents 1
Event | Format-List *


Message              : An account failed to log on.

                       Subject:
                        Security ID:           S-1-5-19
                        Account Name:          LOCAL SERVICE
                        Account Domain:        NT AUTHORITY
                        Logon ID:              0x3E5

                       Logon Type:                     2

                       Account For Which Logon Failed:
                        Security ID:           S-1-0-0
                        Account Name:          [redacted]
                        Account Domain:        -

                       Failure Information:
                        Failure Reason:        Unknown user name or bad password.
                        Status:                0xC000006D
                        Sub Status:            0xC000006A

                       Process Information:
                        Caller Process ID:     0x4ddc
                        Caller Process Name:   C:\Windows\System32\WUDFHost.exe

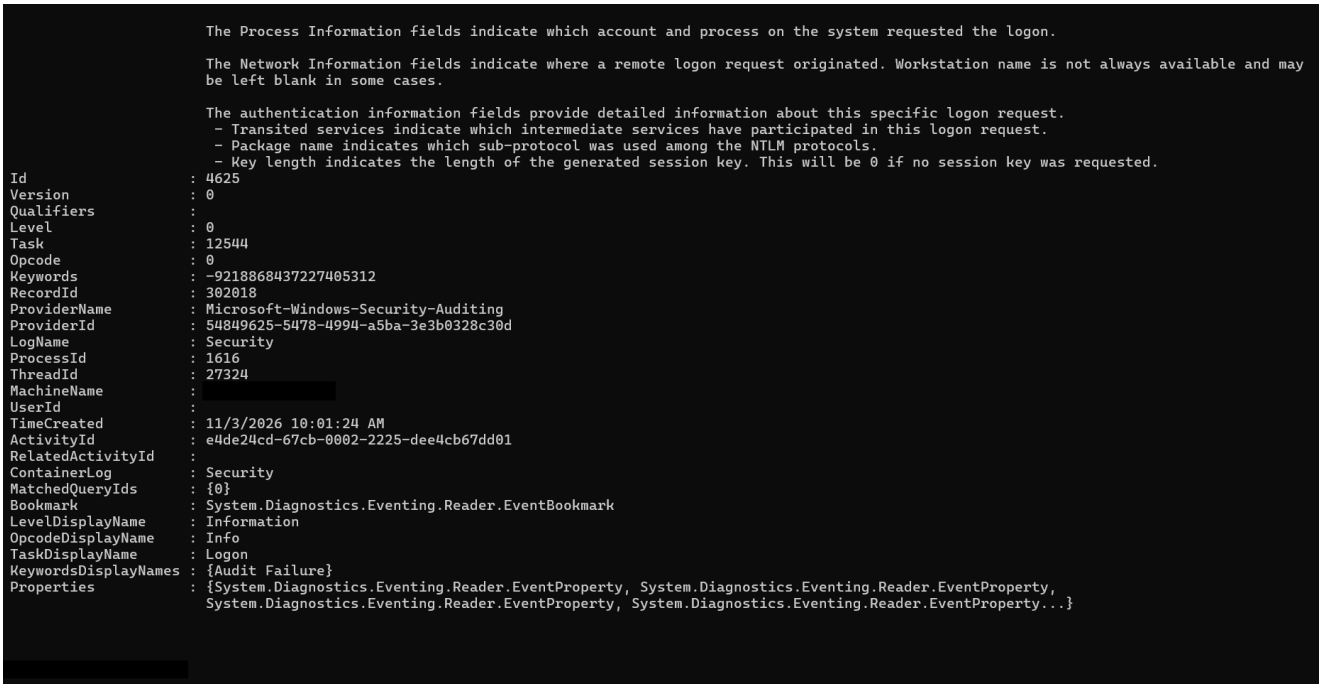
                       Network Information:
                        Workstation Name:      [redacted]
                        Source Network Address: -
                        Source Port:           -

                       Detailed Authentication Information:
                        Logon Process:         Advapi
                        Authentication Package: MICROSOFT_AUTHENTICATION_PACKAGE_V1_0
                        Transited Services:    -
                        Package Name (NTLM only):       -
                        Key Length:            0
```

What it shows: First portion of a failed-logon event detail view.

Explanation: Event ID 4625 records a failed authentication attempt and the detailed view exposes the fields needed for triage. The account, host, failure reason, logon type, process context, and source information should be reviewed together rather than counting failures alone.

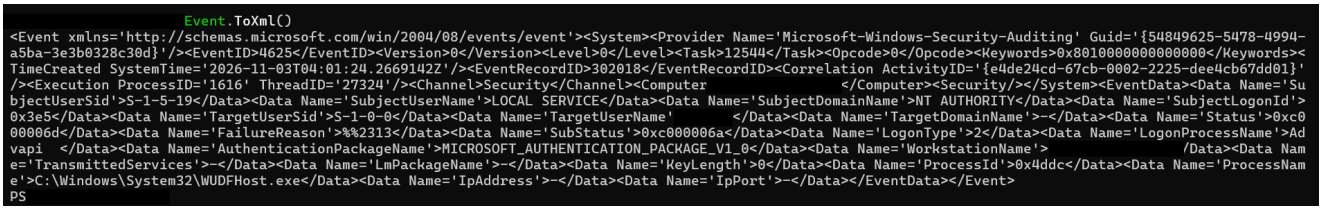
Evidence 09: 09-failed-logon-details-continuation.png



What it shows: Continuation of authentication, process, and network fields.

Explanation: The continuation contains authentication, process, and network-related fields that help distinguish local, network, service, and remote-interactive activity. Missing values, loopback addresses, and system-generated records must be interpreted carefully because they may not represent a remote user.

Evidence 10: 10-failed-logon-event-xml.png



What it shows: Event XML showing named Security event fields.

Explanation: Inspecting the XML verifies the actual Data Name fields present in that event and makes the extraction logic more reliable. This is important because field order and available fields can differ across Event IDs, Windows versions, and event revisions.

Evidence 11: 11-failed-logon-summary-table.png

```
>>     ForEach-Object {
>>         [PSCustomObject]@{
>>             TimeCreated = $_.TimeCreated
>>             TargetUser = $_.Properties[5].Value
>>             SourceIP = $_.Properties[19].Value
>>             LogonType = $_.Properties[8].Value
>>             FailureReason = $_.Properties[13].Value
>>         }
>>     } | Format-Table -AutoSize

TimeCreated            TargetUser SourceIP LogonType FailureReason
-----------            ---------- -------- --------- -------------
11/3/2026 10:01:24 AM [redacted]  -        %%2313    [redacted]
11/3/2026 10:01:23 AM             -        %%2313
11/2/2026 6:45:13 PM              -        %%2313
11/2/2026 8:35:10 AM              -        %%2313
11/2/2026 8:35:10 AM              -        %%2313
11/1/2026 8:30:38 PM              -        %%2313
11/1/2026 2:29:57 PM              -        %%2313
10/31/2026 7:21:58 PM             -        %%2313
10/31/2026 10:38:08 AM            -        %%2313
10/30/2026 11:56:06 PM            -        %%2313
```

What it shows: Selected fields from failed-logon records in a table.

Explanation: The table gives the analyst a compact way to compare repeated failures by time, account, source, and logon type. It supports grouping and prioritization, but the original event record should remain available so that important status, process, and session fields are not lost.

Evidence 12: 12-source-ip-analysis-query.png

```
>>     LogName='Security'
>>     ID=4625
>>     StartTime=(Get-Date).AddDays(-1)
>> } | ForEach-Object {
>>     $_.Properties[19].Value
>> } | Where-Object { $_ -ne '-' -and $_ -ne '' } |
>>     Group-Object |
>>     Sort-Object Count -Descending |
>>     Select-Object -First 5 Name, Count

>>     LogName='Security'
>>     ID=4625
>>     StartTime=(Get-Date).AddDays(-1)
>> } | ForEach-Object {
>>     $_.Properties[19].Value
>> } | Where-Object { $_ -ne '-' -and $_ -ne '' } |
>>     Group-Object |
>>     Sort-Object Count -Descending |
>>     Select-Object -First 5 @{Name='IP Address';Expression={$_.Name}}, @{Name='Failed Attempts';Expression={$_.Count}} |
>>     Format-Table -AutoSize
```

What it shows: Pipeline for grouping failed logons by source address.

Explanation: Grouping failed logons by source address highlights whether activity is concentrated on one source or spread across several sources. A source address may represent NAT, a proxy, a shared system, or synthetic training data, so it is an investigative lead rather than attribution or proof of attack.

Evidence 13: 13-top-targeted-accounts.png

```
                  et-WinEvent -FilterHashtable @{
>>     LogName='Security'
>>     ID=4625
>>     StartTime=(Get-Date).AddDays(-1)
>> } | ForEach-Object {
>>     $_.Properties[5].Value  # TargetUserName
>> } | Where-Object { $_ -ne '-' -and $_ -ne '' } |
>>     Group-Object |
>>     Sort-Object Count -Descending |
>>     Select-Object -First 5 @{Name='Username';Expression={$_.Name}}, @{Name='Failed Attempts';Expression={$_.Count}} |
>>     Format-Table -AutoSize

Username Failed Attempts
-------- ---------------
[redacted]                3
```

What it shows: Failed-logon counts grouped by target account.

Explanation: Grouping by target account identifies which identities received the most failed authentication attempts. The account owner should confirm whether the activity could be explained by a user mistake, a service configuration, approved testing, or an unauthorized access attempt.

Evidence 14: 14-failed-logons-by-type.png

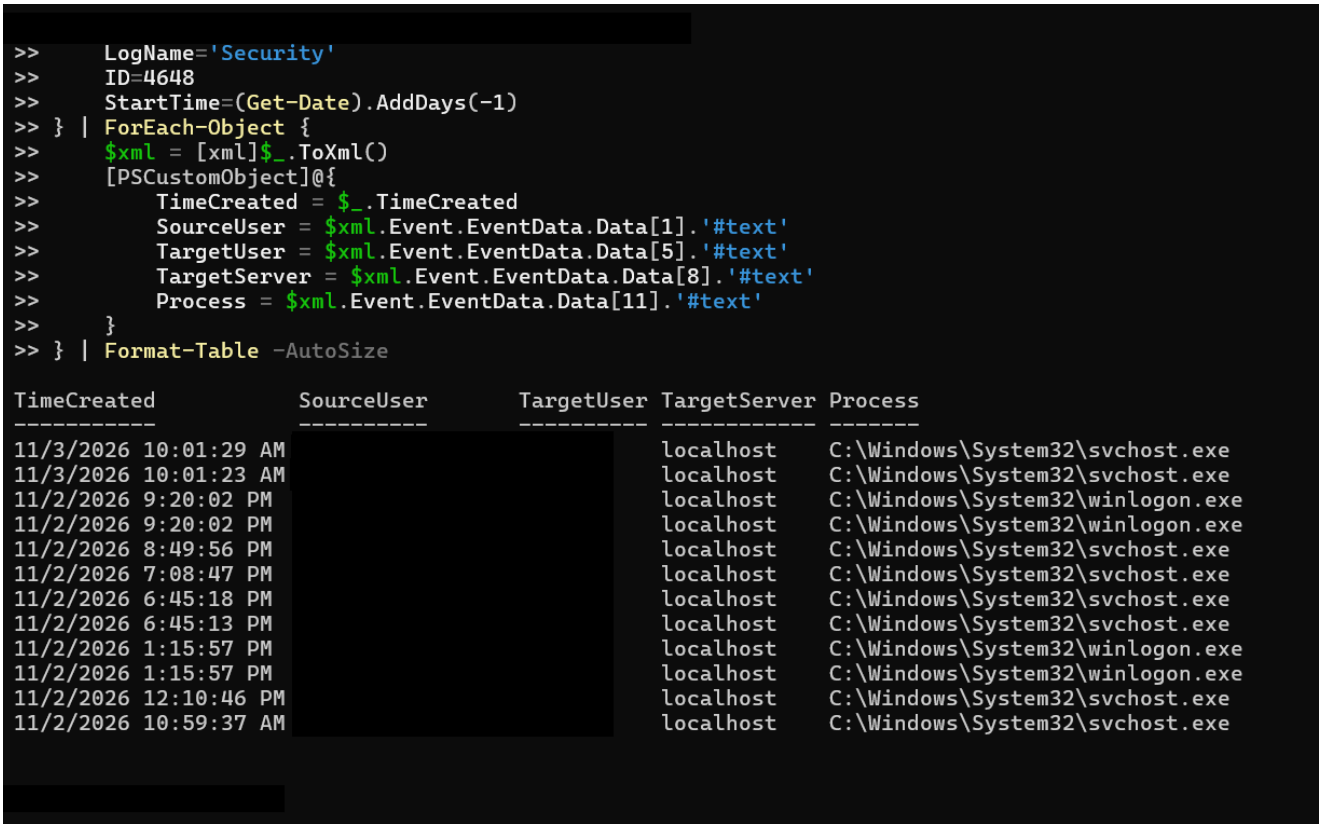
```
>>     LogName='Security'
>>     ID=4625
>>     StartTime=(Get-Date).AddDays(-1)
>> } | ForEach-Object {
>>     $LogonType = $_.Properties[8].Value
>>     switch ($LogonType) {
>>         2 { 'Interactive' }
>>         3 { 'Network' }
>>         4 { 'Batch' }
>>         5 { 'Service' }
>>         7 { 'Unlock' }
>>         8 { 'NetworkCleartext' }
>>         9 { 'NewCredentials' }
>>         10 { 'RemoteInteractive' }
>>         11 { 'CachedInteractive' }
>>         default { "Unknown ($LogonType)" }
>>     }
>> } | Group-Object |
>>     Sort-Object Count -Descending |
>>     Select-Object @{Name='Logon Type';Expression={$_.Name}}, @{Name='Count';Expression={$_.Count}} |
>>     Format-Table -AutoSize

Logon Type       Count
----------       -----
Unknown (%%2313)     3
```

What it shows: Failed-logon grouping by logon type.

Explanation: Logon type helps distinguish interactive, network, batch, service, and remote-interactive patterns. It narrows the investigation and suggests which telemetry to correlate, but it does not independently prove that the activity was unauthorized.

Evidence 15: 15-explicit-credentials-events.png



```
>>     LogName='Security'
>>     ID=4648
>>     StartTime=(Get-Date).AddDays(-1)
>> } | ForEach-Object {
>>     $xml = [xml]$_.ToXml()
>>     [PSCustomObject]@{
>>         TimeCreated = $_.TimeCreated
>>         SourceUser = $xml.Event.EventData.Data[1].'#text'
>>         TargetUser = $xml.Event.EventData.Data[5].'#text'
>>         TargetServer = $xml.Event.EventData.Data[8].'#text'
>>         Process = $xml.Event.EventData.Data[11].'#text'
>>     }
>> } | Format-Table -AutoSize

TimeCreated           SourceUser       TargetUser TargetServer Process
-----------           ----------       ---------- ------------ -------
11/3/2026 10:01:29 AM                             localhost    C:\Windows\System32\svchost.exe
11/3/2026 10:01:23 AM                             localhost    C:\Windows\System32\svchost.exe
11/2/2026 9:20:02 PM                              localhost    C:\Windows\System32\winlogon.exe
11/2/2026 9:20:02 PM                              localhost    C:\Windows\System32\winlogon.exe
11/2/2026 8:49:56 PM                              localhost    C:\Windows\System32\svchost.exe
11/2/2026 7:08:47 PM                              localhost    C:\Windows\System32\svchost.exe
11/2/2026 6:45:18 PM                              localhost    C:\Windows\System32\svchost.exe
11/2/2026 6:45:13 PM                              localhost    C:\Windows\System32\svchost.exe
11/2/2026 1:15:57 PM                              localhost    C:\Windows\System32\winlogon.exe
11/2/2026 1:15:57 PM                              localhost    C:\Windows\System32\winlogon.exe
11/2/2026 12:10:46 PM                             localhost    C:\Windows\System32\svchost.exe
11/2/2026 10:59:37 AM                             localhost    C:\Windows\System32\svchost.exe
```

What it shows: Event ID 4648 review including process and target context.

Explanation: Event ID 4648 indicates that explicit credentials were supplied to a process or connection. This can be normal for backup software, administration, or automation; correlate the initiating process, target account, host, and approved task before treating it as suspicious.

Evidence 16: 16-special-privilege-events.png

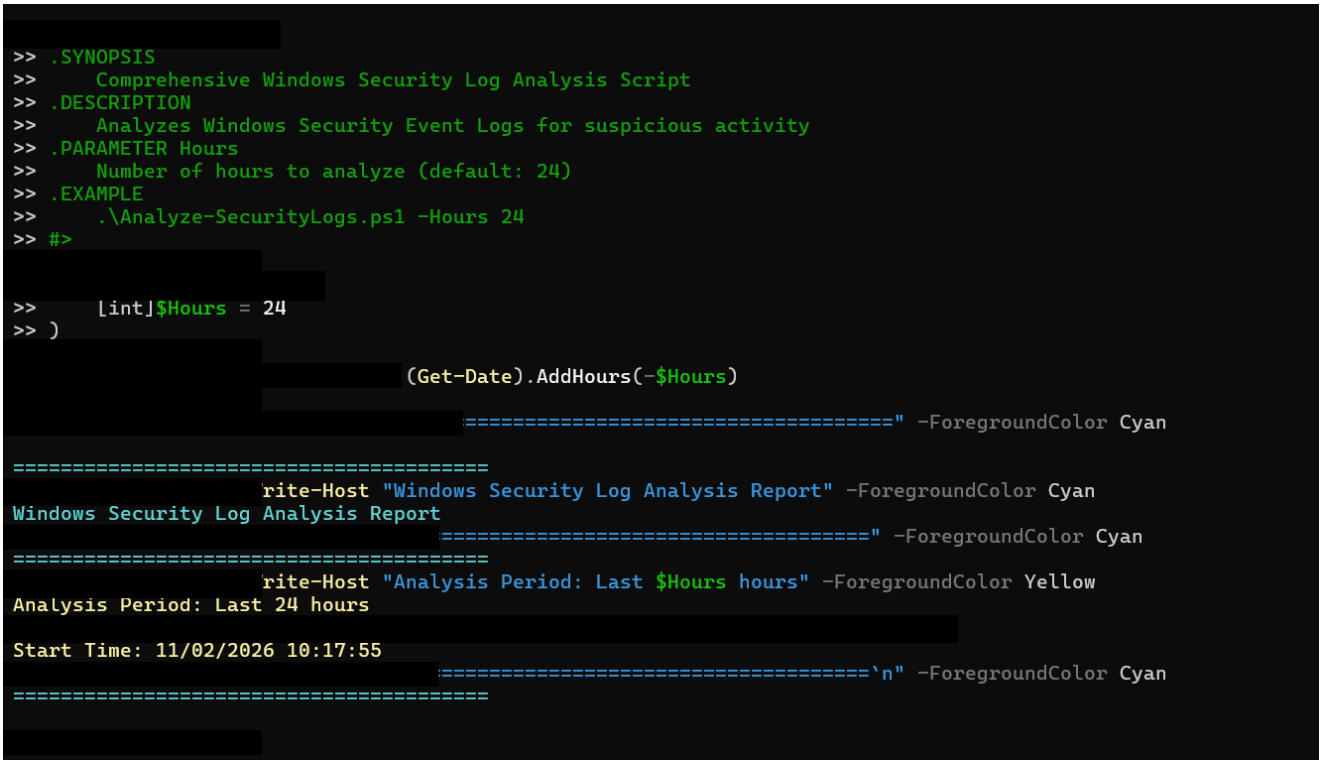
```
>>     LogName='Security'
>>     ID=4672
>>     StartTime=(Get-Date).AddHours(-1)
>> } | ForEach-Object {
>>     $xml = [xml]$_.ToXml()
>>     [PSCustomObject]@{
>>         TimeCreated = $_.TimeCreated
>>         Account = $xml.Event.EventData.Data[1].'#text'
>>         LogonID = $xml.Event.EventData.Data[3].'#text'
>>     }
>> } | Where-Object { $_.Account -notlike '*$' } |  # Filter out system accounts
>>     Format-Table -AutoSize

TimeCreated           Account LogonID
-----------           ------- -------
11/3/2026 10:16:28 AM SYSTEM  0x3e7
11/3/2026 10:16:27 AM SYSTEM  0x3e7
11/3/2026 10:16:25 AM SYSTEM  0x3e7
11/3/2026 10:16:24 AM SYSTEM  0x3e7
11/3/2026 10:14:59 AM SYSTEM  0x3e7
11/3/2026 10:11:55 AM SYSTEM  0x3e7
11/3/2026 10:07:48 AM SYSTEM  0x3e7
11/3/2026 10:07:24 AM SYSTEM  0x3e7
11/3/2026 10:07:21 AM SYSTEM  0x3e7
11/3/2026 10:07:21 AM SYSTEM  0x3e7
11/3/2026 10:06:36 AM SYSTEM  0x3e7
11/3/2026 10:06:26 AM SYSTEM  0x3e7
11/3/2026 10:03:08 AM SYSTEM  0x3e7
11/3/2026 10:02:30 AM SYSTEM  0x3e7
11/3/2026 10:02:05 AM SYSTEM  0x3e7
11/3/2026 10:01:53 AM SYSTEM  0x3e7
11/3/2026 10:01:44 AM SYSTEM  0x3e7
11/3/2026 10:01:37 AM SYSTEM  0x3e7
11/3/2026 10:01:29 AM Lenovo  0x180297db
11/3/2026 10:01:26 AM SYSTEM  0x3e7
11/3/2026 10:01:25 AM SYSTEM  0x3e7
11/3/2026 10:01:24 AM SYSTEM  0x3e7
```

What it shows: Event ID 4672 review by account and logon session.

Explanation: Event ID 4672 indicates that special privileges were assigned to a logon session. Analysts should identify the account, session, host, and related 4624 event, then verify whether the privileged access was expected; the event alone does not establish malicious activity.

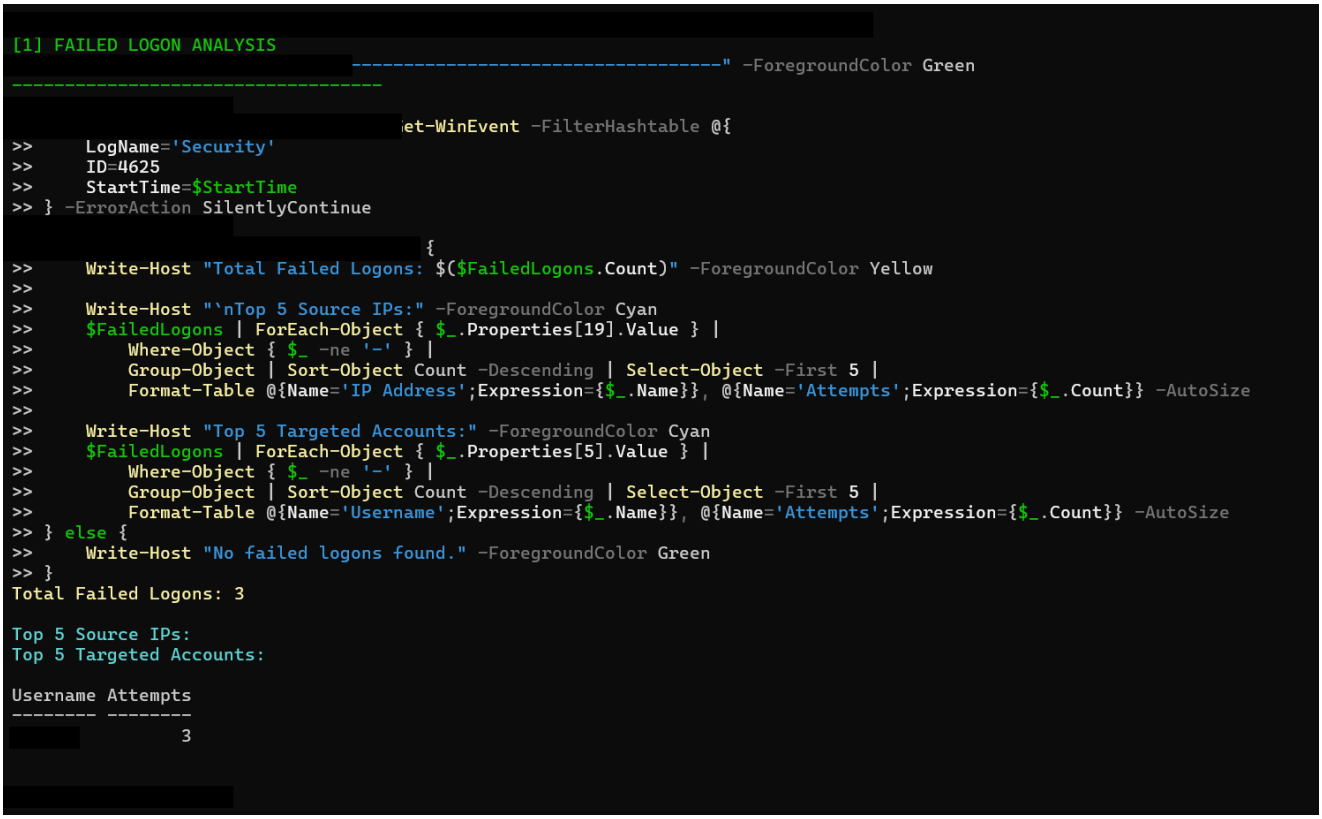
Evidence 17: 17-analysis-script-initialization.png



What it shows: Beginning of the reusable Windows log analysis script.

Explanation: The script initialization shows how the reusable analyzer accepts an input CSV, validates parameters, and applies strict error handling. This creates a consistent offline workflow and avoids querying, modifying, or generating events on a live Windows system.

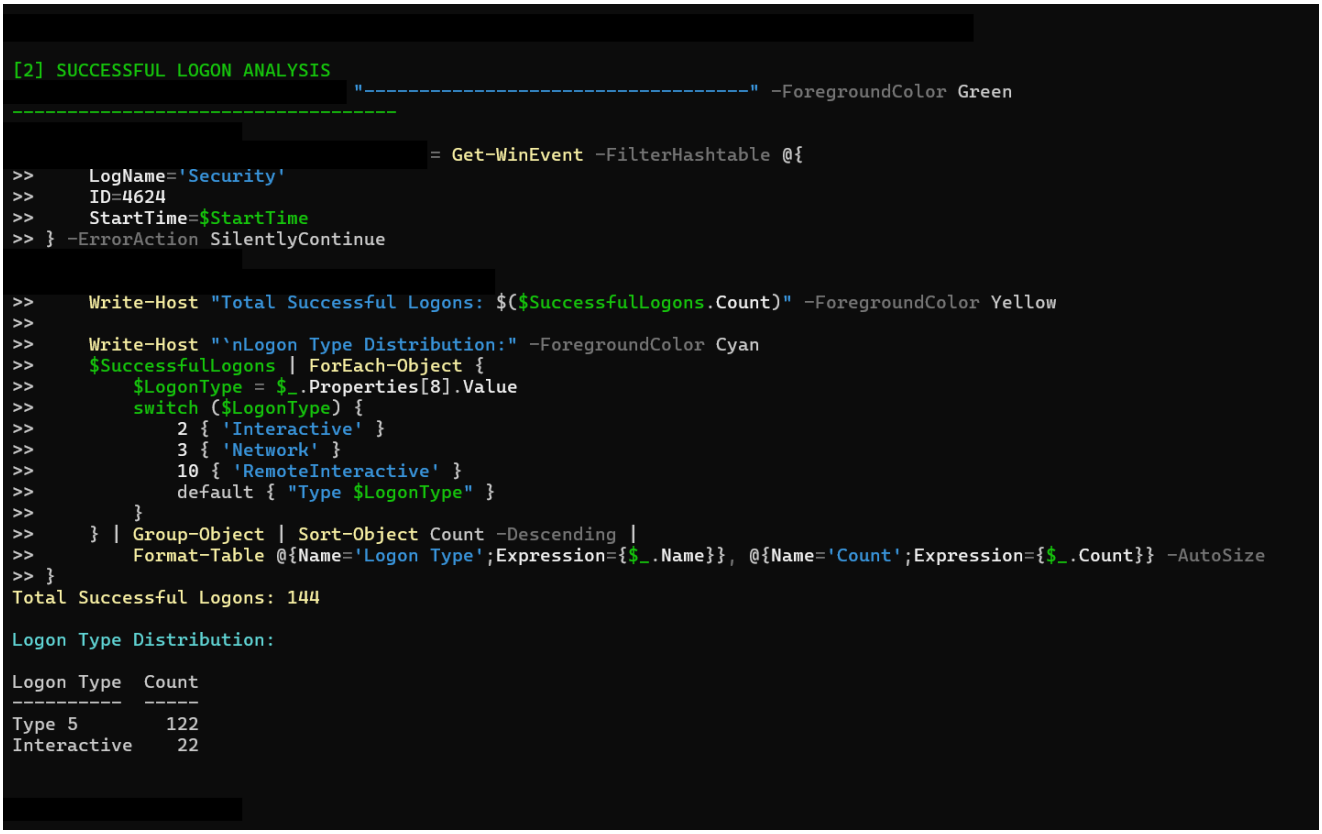
Evidence 18: 18-failed-logon-analysis-output.png



What it shows: Failed-logon analysis section of the script output.

Explanation: This output summarizes the failed-logon portion of the synthetic dataset and makes repeated 4625 activity easy to identify. The summary supports the investigation lead, but the analyst must still correlate the result with successful logons, account ownership, timing, and independent telemetry.

Evidence 19: 19-successful-logon-analysis-output.png



```
[2] SUCCESSFUL LOGON ANALYSIS
                                "----------------------------------" -ForegroundColor Green
----------------------------------

                                = Get-WinEvent -FilterHashtable @{
>>     LogName='Security'
>>     ID=4624
>>     StartTime=$StartTime
>> } -ErrorAction SilentlyContinue

>>     Write-Host "Total Successful Logons: $($SuccessfulLogons.Count)" -ForegroundColor Yellow
>>
>>     Write-Host "`nLogon Type Distribution:" -ForegroundColor Cyan
>>     $SuccessfulLogons | ForEach-Object {
>>         $LogonType = $_.Properties[8].Value
>>         switch ($LogonType) {
>>             2 { 'Interactive' }
>>             3 { 'Network' }
>>             10 { 'RemoteInteractive' }
>>             default { "Type $LogonType" }
>>         }
>>     } | Group-Object | Sort-Object Count -Descending |
>>         Format-Table @{Name='Logon Type';Expression={$_.Name}}, @{Name='Count';Expression={$_.Count}} -AutoSize
>> }
Total Successful Logons: 144

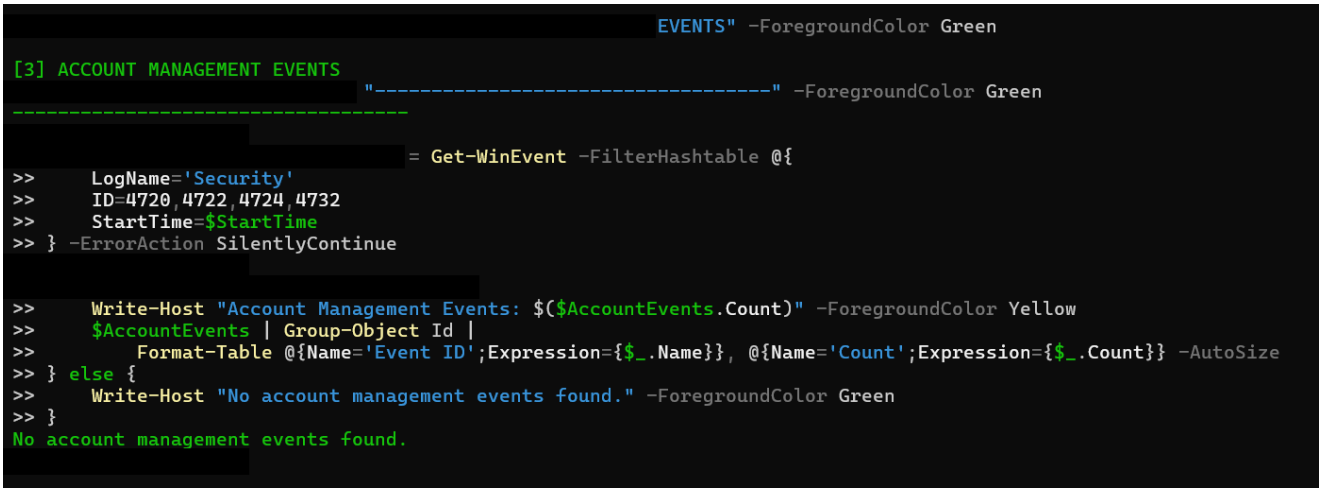
Logon Type Distribution:

Logon Type  Count
----------  -----
Type 5        122
Interactive    22
```

What it shows: Successful-logon counts and logon-type distribution.

Explanation: The successful-logon counts and logon-type distribution show whether an authentication success followed the failure burst and what access method was recorded. Correlation with account ownership, source reputation, host activity, and expected access patterns is required before drawing an incident conclusion.

Evidence 20: 20-account-management-analysis-output.png



```
                                                EVENTS" -ForegroundColor Green

[3] ACCOUNT MANAGEMENT EVENTS
                                "----------------------------------" -ForegroundColor Green
----------------------------------

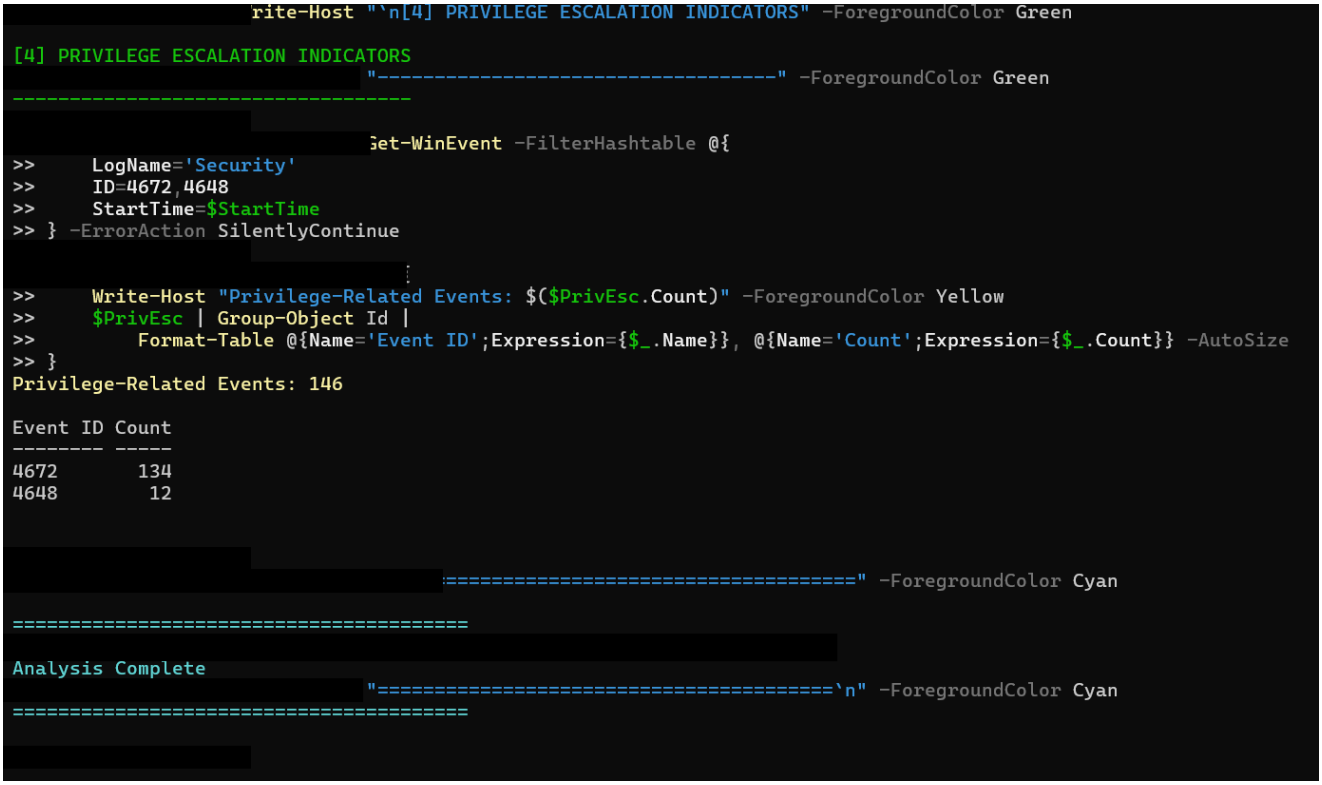
                                = Get-WinEvent -FilterHashtable @{
>>     LogName='Security'
>>     ID=4720,4722,4724,4732
>>     StartTime=$StartTime
>> } -ErrorAction SilentlyContinue

>>     Write-Host "Account Management Events: $($AccountEvents.Count)" -ForegroundColor Yellow
>>     $AccountEvents | Group-Object Id |
>>         Format-Table @{Name='Event ID';Expression={$_.Name}}, @{Name='Count';Expression={$_.Count}} -AutoSize
>> } else {
>>     Write-Host "No account management events found." -ForegroundColor Green
>> }
No account management events found.
```

What it shows: Account-management event review.

Explanation: Account-management events such as user creation should be checked against change tickets, administrator activity, and the intended system owner. A new account can be legitimate helpdesk or onboarding work, so the event is a review point rather than proof of persistence.

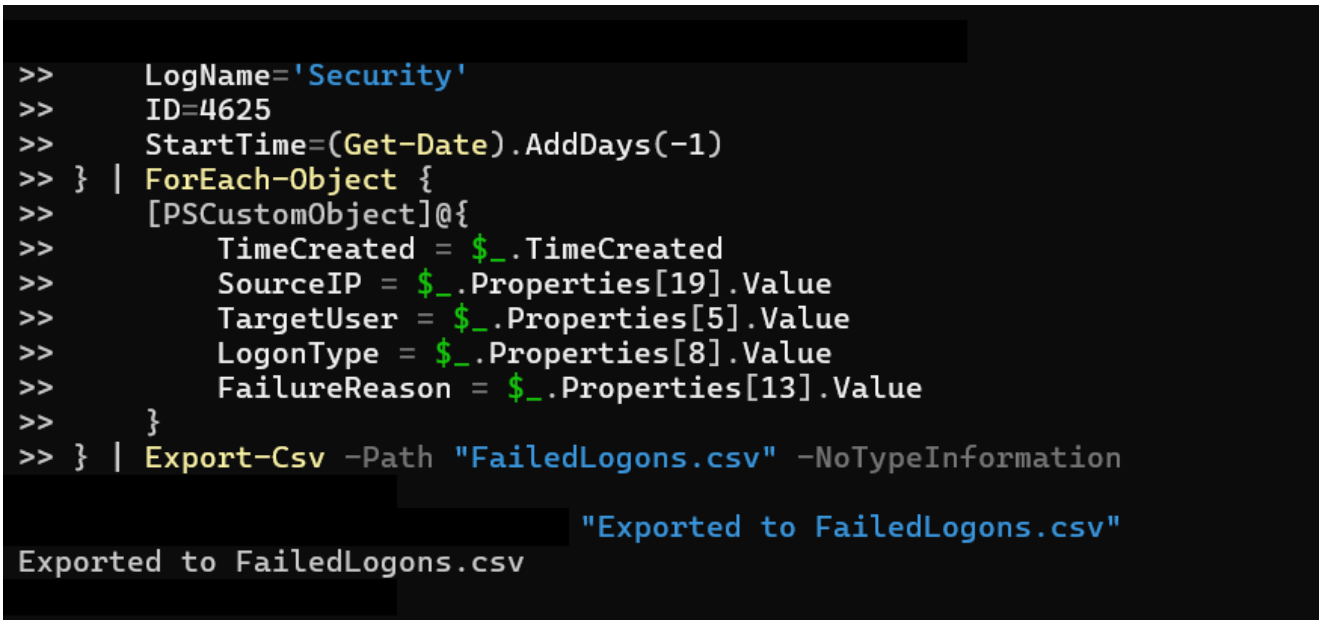
Evidence 21: 21-privilege-indicator-analysis-output.png



What it shows: Counts for privilege-related events.

Explanation: Privilege-related counts identify events that may affect the security posture of the host. Review the account, logon session, process, timing, and approved administrative work together; a count without context cannot distinguish routine maintenance from misuse.

Evidence 22: 22-csv-export-command.png



What it shows: Export of selected failed-logon fields to CSV.

Explanation: Exporting selected fields to CSV creates a structured, portable evidence set for review and reporting. The export should preserve UTC timestamps and enough source context to reproduce the grouping logic, while avoiding unnecessary sensitive data.

Evidence 23: 23-html-report-export-command.png

```
Exported to FailedLogons.csv

>>     LogName='Security'
>>     ID=4625
>>     StartTime=(Get-Date).AddDays(-1)
>> } | ForEach-Object {
>>     [PSCustomObject]@{
>>         TimeCreated = $_.TimeCreated
>>         SourceIP = $_.Properties[19].Value
>>         TargetUser = $_.Properties[5].Value
>>         LogonType = $_.Properties[8].Value
>>     }
>> }

                    HTML = $FailedLogons | ConvertTo-Html -Title "Failed Logon Report" -PreContent "<h1>Failed Logon Analysis</h1><p>Generated: $(Get-Date)
</p>"

                    HTML | Out-File "FailedLogons.html"

Exported to FailedLogons.html
```

What it shows: Creation of an HTML summary report.

Explanation: The HTML export turns the analysis into a human-readable report suitable for analyst handoff and review. A useful report records observations, interpretation, confidence, limitations, and next steps instead of presenting suspicious-looking counts as confirmed incidents.

Evidence 24: 24-offline-analyzer-synthetic-demo.png

```
SYNTHETIC TRAINING DATA - OFFLINE DEMONSTRATION
PS> .\tools\Analyze-SecurityEvents.ps1 -InputPath .\data\windows-security-events.csv
   -OutputPath .\output\windows-security-analysis.md

Report written to: .\output\windows-security-analysis.md
Events analyzed: 12; failed logons: 6; successful logons: 1
# Windows Security Event Analysis

> Offline analysis of the supplied CSV. The reference time is the newest event in the dataset; the window is relative to that event, not the current

 - **Input:** `windows-security-events.csv`
 - **Window:** 2026-11-02 05:47:00Z to 2026-11-03 05:47:00Z
 - **Records in window:** 12
 - **Failed logons (4625):** 6
 - **Successful logons (4624):** 1

## Top sources among failed logons

- Source IP: 203.0.113.42 - 6 event(s)

## Most-targeted accounts

- Account: alex.lee - 6 event(s)

## Other review-worthy event counts

- Event ID 4648: 1
- Event ID 4672: 1
- Event ID 4720: 1
- Event ID 4732: 1
- Event ID 5140: 1

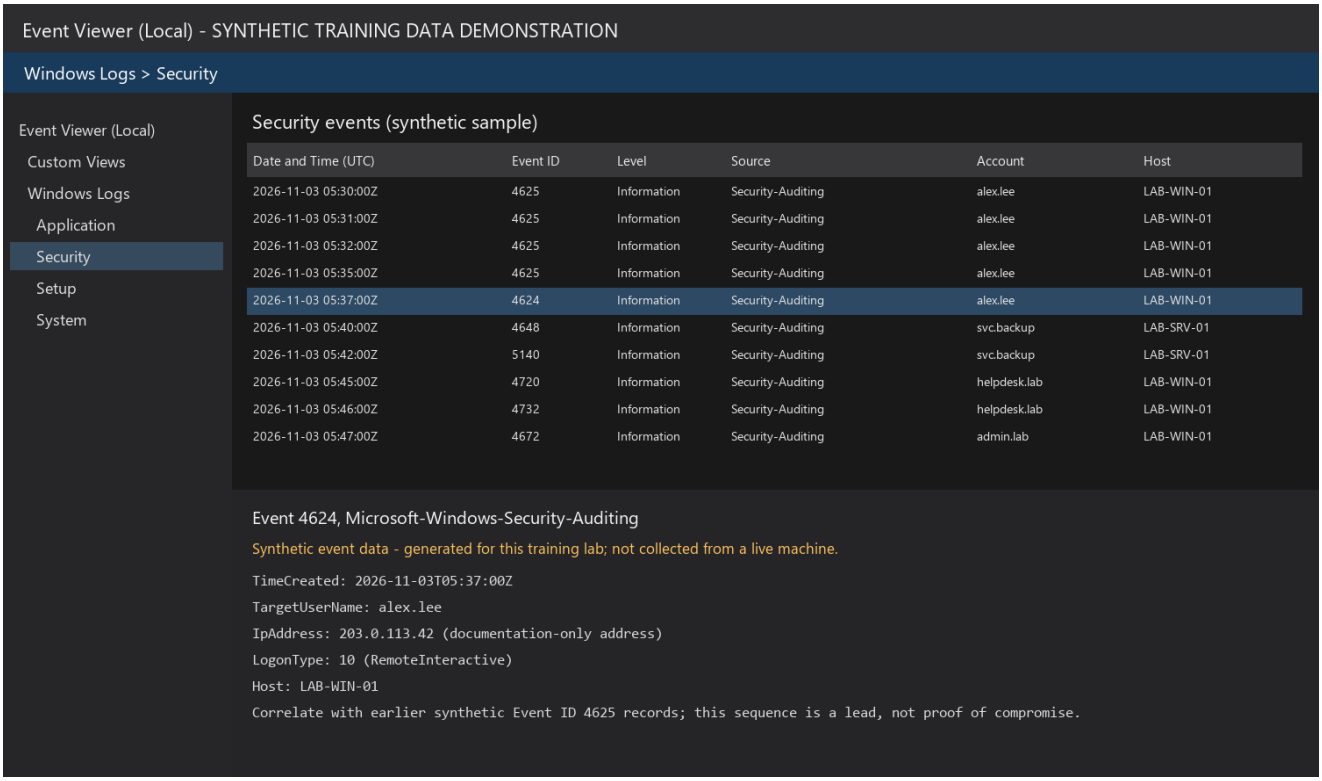
## Analyst notes

- A repeated failure or a success after failures is an investigative lead, not proof of compromise.
- Correlate account, source, host, logon type, approved activity, and relevant endpoint/network evidence.
- This synthetic dataset does not establish real-world attribution or intent.
```

What it shows: Synthetic offline analyzer output using the bundled CSV.

Explanation: This is the safe, repeatable demonstration using the repository-provided synthetic CSV rather than a live Security log. It allows the learner to validate parsing, counting, grouping, and report generation without administrator rights or interaction with a real system.

Evidence 25: 25-event-viewer-synthetic-demo.png



What it shows: Synthetic Event Viewer-style demonstration; not a live system screenshot.

Explanation: This illustrates how synthetic event records may be displayed in an Event Viewer-style layout for training. It is not a capture from a live machine and must not be used as production evidence or as proof of a real user, host, or incident.

10. Completion checklist

This report covers the offline dataset, event-ID interpretation, failed/successful logon correlation, named-field methodology, limitations, recommendations, and the complete screenshot catalog. Before publishing, verify that any additional screenshots or logs are synthetic or properly redacted.

Requirement	Status
Offline analyzer and dataset reviewed	Complete
Failed logons correlated with other event types	Complete
Alternative explanations and limitations documented	Complete
Evidence catalog included	Complete
Real logs and credentials excluded from this report	Based on repository screenshot guide